

Ordinary Least Squares and the Normal Equations

These notes develop the normal equations for ordinary least squares (OLS). The goal is to connect derivative-based optimization from calculus with the matrix notation used in regression.

Learning goals

By the end of these notes, you should be able to write the residual sum of squares in matrix form, apply the matrix-derivative rules needed for OLS, derive the normal equations, verify that their solution minimizes the loss, and interpret them geometrically.

1. Setup and notation

We observe n responses and use p coefficients, including an intercept coefficient when the model has an intercept.

Symbol	Dimensions	Meaning
X	$n \times p$	Design matrix; row i is $x_i^\top$
y	$n \times 1$	Response vector
β	$p \times 1$	Candidate coefficient vector
$X\beta$	_____	Fitted-value vector
$r = y - X\beta$	_____	Residual vector

The linear model is $y = X\beta + \varepsilon$. For a candidate β ,

$$\hat{y} = X\beta, \quad r = y - X\beta.$$

If the model contains an intercept, the first column of X is $\mathbf{1}$.

Quick dimension check

Complete each dimension. If a product is not defined, explain why.

Expression	Dimensions
$X\beta$	_____
$X^\top y$	_____
$X^\top X$	_____
$\beta^\top X^\top y$	_____
$XX^\top$	_____

2. From scalar to matrix least squares

In scalar notation, the residual sum of squares is

$$\text{RSS}(\beta) = \sum_{i=1}^n (y_i - x_i^\top \beta)^2.$$

The same objective in matrix notation is

$$L(\beta) = \|y - X\beta\|_2^2 = (y - X\beta)^\top (y - X\beta).$$

OLS chooses $\hat{\beta}$ to minimize $L(\beta)$.

What dimensions must $\nabla_\beta L(\beta)$ have? Why?

Think first

- 1. Why is $L(\beta)$ always nonnegative?
- 2. Why does minimizing $L(\beta)/n$ give the same $\hat{\beta}$?
- 3. Does defining this problem require normally distributed errors?

3. Matrix-derivative toolbox

Let $\mathbf{b} \in \mathbb{R}^p$ be the column vector with respect to which we differentiate. In this toolbox, **bold lowercase letters denote vectors**, ordinary lowercase letters denote scalar constants, and uppercase letters denote constant matrices. Thus $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{d}$ are vectors; c is a scalar constant; and A and B are constant matrices. We use the **column-gradient convention**, so $\nabla_{\mathbf{b}} f$ is $p \times 1$ when f is scalar-valued. Each rule assumes compatible dimensions; take $A \in \mathbb{R}^{p \times p}$ and, in rules containing B , take $B \in \mathbb{R}^{m \times p}$ and $\mathbf{a}, \mathbf{d} \in \mathbb{R}^m$.

Function $f(\mathbf{b})$	Gradient $\nabla_{\mathbf{b}} f(\mathbf{b})$	Note
c	0	Constant rule
$\mathbf{a}^\top \mathbf{b}$	$\mathbf{a}$	Linear form

Function $f(\mathbf{b})$	Gradient $\nabla_{\mathbf{b}} f(\mathbf{b})$	Note
$\mathbf{b}^\top \mathbf{a}$	$\mathbf{a}$	Same scalar as $\mathbf{a}^\top \mathbf{b}$
$\mathbf{d}^\top B \mathbf{b}$	$B^\top \mathbf{d}$	Matrix inside a linear form
$\mathbf{b}^\top A \mathbf{b}$	$(A + A^\top) \mathbf{b}$	General quadratic form
$\mathbf{b}^\top A \mathbf{b}$, if $A = A^\top$	$2A \mathbf{b}$	Symmetric quadratic form

4. Guided derivation of the normal equations

Step 1: Expand the loss

Begin with

$$L(\beta) = (\mathbf{y} - X\beta)^\top (\mathbf{y} - X\beta).$$

Recall that $(AB)^\top = B^\top A^\top$.

Expand the product. Show all four terms and check that every final term is a scalar.

$$\begin{aligned} L(\beta) &= (\mathbf{y} - X\beta)^\top (\mathbf{y} - X\beta) \\ &= \end{aligned}$$

Step 2: Recognize equal scalar terms

The two cross-product terms are transposes of one another. Complete the argument:

$$\mathbf{y}^\top X\beta \text{ has dimensions } \implies \mathbf{y}^\top X\beta = (\mathbf{y}^\top X\beta)^\top = \quad .$$

Use this fact to write the expanded loss with three terms:

$$L(\beta) =$$

Step 3: Differentiate term by term

Use the toolbox from Section 3. Record the rule you use before simplifying.

$$\nabla_{\beta} L(\beta) = \nabla_{\beta}(y^{\top} y) - 2\nabla_{\beta}(\beta^{\top} X^{\top} y) + \nabla_{\beta}(\beta^{\top} X^{\top} X \beta)$$

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Step 4: Apply the first-order condition

At an unconstrained differentiable minimum, the gradient must equal zero. Set your result from Step 3 equal to zero:

$$= 0.$$

These are called **normal equations** because they state that the residual is normal (perpendicular) to the columns of the design matrix—not because they require normally distributed errors.

Step 5: Solve for $\hat{\beta}$

If X has full column rank, then $X^\top X$ is invertible. Starting from the normal equations, solve for $\hat{\beta}$.

$\hat{\beta} =$

Where did the full-column-rank assumption enter the derivation?

5. Why the critical point is a minimum

A zero gradient identifies a critical point, but not whether it is a minimum, maximum, or saddle point.

Differentiate the gradient once more:

$$H(\beta) = \nabla_\beta^2 L(\beta) =$$

For any nonzero $v \in \mathbb{R}^p$,

$$v^\top X^\top X v = (Xv)^\top (Xv) =$$

Complete the reasoning:

- 1. This quantity is always.
- 2. If X has full column rank and $v \neq 0$, then Xv is.
- 3. Therefore, is $X^\top X$ positive semidefinite or positive definite?
- 4. What does this imply about $L(\beta)$ and its critical point?

What changes when X is rank deficient? Can a minimum still exist, and must its coefficient vector be unique?

6. Derivation assumptions versus inference assumptions

Define/derive OLS	Obtain a unique inverse-form solution	Conduct statistical inference
Fitted values have the form $X\beta$	X has full column rank	$\mathbb{E}[\varepsilon \mid X] = 0$
Entries of X and y are finite	Equivalently, $X^\top X$ is invertible	$\text{Var}(\varepsilon \mid X) = \sigma^2 I_n$
Full rank is not required to state the normal equations		Normal errors are additionally used for exact finite-sample inference

Does deriving the normal equations require normal errors? Explain.

No. Normal errors are not needed to define the least-squares criterion, differentiate it, obtain the normal equations, or calculate an OLS solution. They are relevant only to particular inferential results.

7. Small example: you finish the calculation

Suppose

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Part A: Build the normal equations

$$X^{\top} X = \begin{bmatrix} & \\ & \end{bmatrix}, \quad X^{\top} y = \begin{bmatrix} \\ \end{bmatrix}.$$

Write the resulting system:

$$\left\{ \begin{array}{l} \\ \end{array} \right.$$

Part B: Solve and check

$$\hat{\beta} = \begin{bmatrix} \\ \end{bmatrix}, \quad \hat{y} = \begin{bmatrix} \\ \\ \end{bmatrix}, \quad \hat{r} = \begin{bmatrix} \\ \\ \end{bmatrix}.$$

Verify:

$$X^{\top} \hat{r} = , \quad \mathbf{1}^{\top} \hat{r} = .$$

8. Wrap-up and computation

The formula involving $(X^{\top}X)^{-1}$ is valuable for understanding the derivation. In numerical work, software generally uses a QR decomposition or singular value decomposition (SVD) instead of explicitly forming the inverse. These approaches are more stable when predictors are highly correlated or have very different scales.